The Impact of the 2008 Financial Crisis on Youth Employment in Italy: A Difference-in-Differences Analysis

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Abstract

This paper describes causal inference research on the 2008 Global Financial Crisis and its effects on employment levels of the young population of Italy using model frameworks like DID (difference-in-difference). The data used in the paper is the public EU Labor Force Survey (EU-LFS). The data provide aggregated survey microdata with weighted coefficients, collected from 2004 to 2013 in both yearly and quarterly format. For analysis, survey data from the pre-crisis period (2004–2007) and the post-crisis period (2009–2010) are used. The main target is the employment outcomes of young workers (the age group of 20-year-olds) relatively to prime-age workers (the age groups of 32- and 47-year-olds). The results show a reduction in employment for young workers by 3.8 percent compared to prime-age workers. This effect is robust to the inclusion of gender, alternative specifications, and placebo tests. In the results, the pre-crisis trend is observed and analyzed using event study estimates and alternative models. The event study confirms parallel pre-crisis trends and shows a significant negative effect from 2009 to 2010. Additionally, there is evidence of a reduction in weekly working hours of 0.5 hours and a rise in part-time employment among younger population. These findings show the disproportional impact of the Global Crisis on the vulnerable - in our case, the younger population of Italy - and emphasize the importance of youth employment policies during economic recessions.

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1 Introduction

The 2008 Global Financial Crisis is undeniably the most severe economic downturn at the beginning of the 21st century. In Italy, this event caused very drastic changes in the economic activity with GDP falling by over 5% and the labor market.

"Job losses have obviously been the most significant social impact of the crisis... The unemployment rate remained almost stable during the initial period of the downturn but increased dramatically in 2009, with further increases expected in 2010. Regarding the labor market, ...inactivity rates have increased, probably because of discouragement, and the disincentives for the long-term unemployed to search for jobs. Moreover, the rise in unemployment has been concentrated in certain categories (younger people)... Finally, the crisis has caused a rise in part-time employment accepted by workers seeking to keep their jobs despite falls in production" (1)

Using a difference-in-differences framework that compares young workers to prime-age workers before and after the crisis, it is estimated that the crisis reduced employment probability for young workers by 3.8 percent more than for prime-age workers. This effect

is robust to the inclusion of individual controls and alternative specifications. Moreover, average weekly hours decreased, while part-time employment increased.

This paper is an analysis of youth labor market outcomes during recessions and provides evidence on the heterogeneous effects of macroeconomic shocks.

2 Data and Empirical Strategy

2.1 Data Source and Sample Construction

The microdata is obtained from the EU Labour Force Survey (LFS), for the years 2004-2010.

> "The EU-LFS is a large household sample survey providing quarterly results on labour participation of people aged 15 and over and on people outside the labour force. The survey covers persons aged 15 years and over who live in private households". (2)

Table 1 presents descriptive statistics by group and period. Before the crisis (2006-2007), employment among young workers was 78.9%, compared to 94.4% for prime-age workers. After the crisis, young workers experienced a 5.5 percent decline, while prime-age workers only 1.6 percent decline in comparison. This difference is the focus of the analysis.

Table 1: Descriptive Statistics by Group and Period

	Prime-age (30-50)		Young (15-24)	
	2006-2007	2009-2010	2006-2007	2009-2010
Employment Rate	0.944	0.928	0.789	0.734
Female	0.415	0.422	0.405	0.401
Age	39.3	39.8	20.0	20.0
Part-time	0.132	0.143	0.181	0.227
Observations	417,895	394,576	41,793	34,338
Weighted N	39,490	39,564	3,754	3,404

Notes: Employment rates and proportions are weighted using LFS sampling weights. Observations are unweighted counts; Weighted N is the sum of weights (in thousands).

2.2 Empirical Strategy

The base difference-in-differences model:

$$Y_{it} = \beta_0 + \beta_1 \text{Treat}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Treat}_i \times \text{Post}_t) + \mathbf{X}_{it}\gamma + \varepsilon_{it} \tag{1}$$

where Y_{it} is the employment outcome for individual i at time t, Treat_i is an indicator equal to 1 for young workers and 0 for prime-age workers, Post_t is an indicator equal to

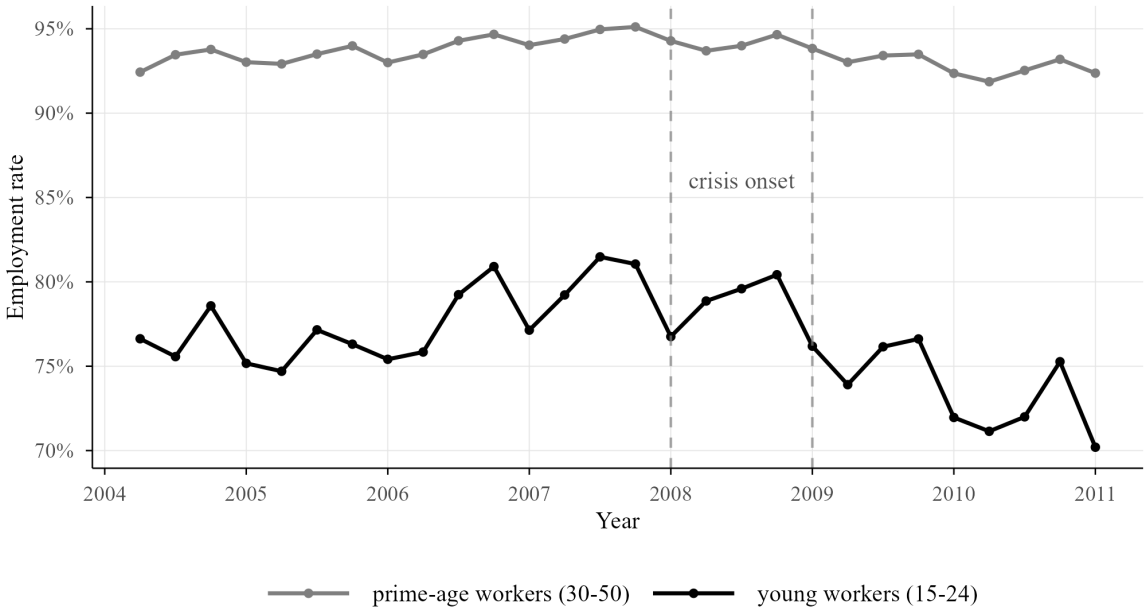

Figure 1: Employment trends by age group, Italy 2006-2010

1 for the post-crisis period (2009-2010) and 0 for the pre-crisis period (2006-2007), and $\mathbf{X}_{it}$ is a vector of individual controls including sex, age, education, and part-time status. Unfortunately, a significant number of values provided in the original data are coded as 9 or 99 to indicate "not applicable". Furthermore, the distributions of some demographic attributes were heavily skewed. Hence, attributes like nationality or education level were not used in the analysis. The coefficient of interest is β_3, which estimates the differential effect of the crisis on young workers compared to prime-age workers.

2008 was excluded from the analysis as it represents a transition year. All regressions use normalized sampling weights to ensure representativeness and heteroskedasticity-robust standard errors.

2.3 Parallel Trends Assumption

The difference-in-differences estimator is only valid if we assume that employment trends for workers and prime-age workers would have been the same without the crisis. This means that the employment trends for workers and prime-age workers should have moved in the same direction.

Figure 1 displays employment trends for both groups from 2004 to 2010. It shows that the employment trends for workers and prime-age workers moved in the same direction from 2004 to 2007 before the crisis. However, the employment rate for the young population drops down after 2008.

3 Main Results

3.1 Basic Difference-in-Differences Estimates

The simplest estimation of the basic difference-in-differences model without any controls. The model is specified as:

$$\text{Employed}_{it} = \beta_0 + \beta_1 \text{Treat}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Treat}_i \times \text{Post}_t) + \varepsilon_{it} \tag{2}$$

Column (1) of Table 2 presents the results. The coefficient on the interaction term (Young × Post-crisis) is -0.039 and statistically significant at the 1% level, indicating that the crisis reduced employment probability for young workers by 3.9 percent more than for prime-age workers.

Next adding controls for sex and age to the model. This is to account for differences, in the makeup of the groups. The model is specified as:

$$\text{Employed}_{it} = \beta_0 + \beta_1 \text{Treat}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Treat}_i \times \text{Post}_t) + \gamma_1 \text{Female}_i + \gamma_2 \text{Age}_i + \varepsilon_{it} \tag{3}$$

Column (2) shows that the DID coefficient remains stable at -0.038, implying that compositional changes do not alter the result. The control variables show expected results: women have lower employment rates (-2.9 percent), and employment increases with age (0.3 percent per year).

Table 2: Main Difference-in-Differences Estimates

	(1) Basic DID	(2) + Demographics
Young (15-24)	-0.155^{***}	-0.104^{***}
	(0.003)	(0.003)
Post-crisis	-0.016^{***}	-0.017^{***}
	(0.001)	(0.001)
Young × Post-crisis	-0.039^{***}	-0.038^{***}
	(0.004)	(0.004)
Female		-0.029^{***}
		(0.001)
Age		0.003^{***}
		(0.000)
Observations	888,602	
R-squared	0.041	

Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Dependent variable: employed (0/1). Heteroskedasticity-robust standard errors. Sampling weights applied.

3.2 Alternative Outcomes: Hours and Part-time Employment

To examine whether the crisis affected the intensive margin of employment and job quality, two additional outcomes: hours worked and part-time employment were analyzed. For these analyses, only employed workers are being considered.

The model for hours worked is specified as:

$$\text{Hours}_{it} = \beta_0 + \beta_1 \text{Treat}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Treat}_i \times \text{Post}_t) + \gamma_1 \text{Female}_i + \gamma_2 \text{Age}_i + \gamma_3 \text{Part-time}_i + \varepsilon_{it} \quad (4)$$

Column (3) of Table 3 depicts the results. The coefficient on Young × Post-crisis is -0.506 (p ¡ 0.001), indicating that young workers lost one half-hour per week compared to prime-age workers.

The model for part-time employment is:

$$\text{Part-time}_{it} = \beta_0 + \beta_1 \text{Treat}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Treat}_i \times \text{Post}_t) + \gamma_1 \text{Female}_i + \gamma_2 \text{Age}_i + \varepsilon_{it} \quad (5)$$

Column (4) shows that the coefficient is 0.036 (p ¡ 0.001), suggesting that young workers became 3.6 percentage points more likely to work part-time compared to prime-age workers. These results show that not only employment rates were reduced, but for the young individuals remained employed the change in working contracts, in this case, from full-time to part-time job is shown.

Table 3: Alternative Outcomes: Hours and Part-time Employment

	(3) Hours Worked	(4) Part-time
Young (15-24)	-0.710^{***}	0.032^{***}
	(0.071)	(0.003)
Post-crisis	0.849^{***}	0.010^{***}
	(0.026)	(0.001)
Young × Post-crisis	-0.506^{***}	0.036^{***}
	(0.082)	(0.004)
Female	-3.093^{***}	0.238^{***}
	(0.027)	(0.001)
Age	-0.031^{***}	-0.001^{***}
	(0.002)	(0.000)
Part-time	-17.456^{***}	
	(0.034)	
Observations	762,926	
R-squared	0.352	

Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Dependent variable: usual hours worked (model 3); part-time indicator (model 4).
Sample restricted to employed workers.

3.3 Triple Difference: Heterogeneity by Gender

To understand whether the crisis effect differs by gender, the triple difference model that interacts the treatment with gender is specified as:

$$\begin{aligned}\text{Employed}_{it} = \beta_0 &+ \beta_1\text{Treat}_i + \beta_2\text{Post}_t + \beta_3\text{Female}_i \\ &+ \beta_4(\text{Treat}_i \times \text{Post}_t) + \beta_5(\text{Treat}_i \times \text{Female}_i) \\ &+ \beta_6(\text{Post}_t \times \text{Female}_i) + \beta_7(\text{Treat}_i \times \text{Post}_t \times \text{Female}_i) \\ &+ \gamma_1\text{Age}_i + \varepsilon_{it}\end{aligned} \tag{6}$$

The coefficient of interest is β_7, which estimates the differential effect of the crisis on young women relative to young men.

Column (5) of Table 4 depicts the results. The triple interaction coefficient (Young × Post × Female) is 0.010 and statistically insignificant (p ¿ 0.1), indicating that the difference of the crisis shock on employment between young men and young women is insignificant.

3.4 Placebo Test: Testing Parallel Trends

The placebo model is used to test the pre-crisis parallel trends. This model compares 2004-2005 and 2006-2007 periods. The model is specified as:

$$\begin{aligned}\text{Employed}_{it} =&\beta_0 + \beta_1\text{Treat}_i + \beta_2\text{PlaceboPost}_t \\ &+ \beta_3(\text{Treat}_i \times \text{PlaceboPost}_t) + \gamma_1\text{Female}_i + \gamma_2\text{Age}_i + \varepsilon_{it}\end{aligned} \tag{7}$$

where PlaceboPost_t equals 1 for years 2006-2007 and 0 for years 2004-2005.

Column (6) of Table 4 shows that the coefficient on the placebo interaction term is 0.022 and statistically significant (p ¡ 0.001), indicating the presence of a modest pre-trend. For this reason further in the paper the robustness check model was executed and this pre-trend is not statistically significant when controlling for a linear time trend.

4 Robustness Checks

4.1 Alternative Specifications

Table 4 presents additional robustness checks. Column (1) reproduces the main specification with 2006-2007 as the pre-crisis period (coefficient -0.038). Column (2) uses only 2007 as the pre-crisis period, yielding a slightly larger coefficient of -0.043, suggesting the main estimate is conservative.

Column (3) includes a linear time trend interacted with the treatment to control for pre-existing trends:

$$\begin{aligned}\text{Employed}_{it} =& \beta_0 + \beta_1 \text{Treat}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Treat}_i \times \text{Post}_t) \\ & \beta_4 \text{Time}_t + \beta_5 (\text{Treat}_i \times \text{Time}_t) + \gamma_1 \text{Female}_i + \gamma_2 \text{Age}_i + \varepsilon_{it}\end{aligned} \tag{8}$$

The coefficient on the interaction term (treat × time_trend) is statistically insignificant (-0.004, $p < 0.1$), indicating that young and prime-age workers did not follow significantly different employment trends before the crisis and the parallel trends assumption is valid. The DID coefficient remains negative and statistically significant at -0.025 ($p < 0.05$).

Table 4: Robustness Checks

	(5) Triple Diff	(6) Placebo Test	(7) With Trend
Young (15-24)	-0.094^{***}	-0.128^{***}	-0.094^{***}
	(0.003)	(0.004)	(0.010)
Post-crisis	-0.020^{***}		-0.016^{***}
	(0.001)		(0.002)
Young × Post-crisis	-0.042^{***}		-0.025^{*}
	(0.005)		(0.012)
Female	-0.031^{***}	-0.036^{***}	-0.029^{***}
	(0.001)	(0.001)	(0.001)
Age	0.003^{***}	0.003^{***}	0.003^{***}
	(0.000)	(0.000)	(0.000)
Young × Female	-0.025^{***}		
	(0.005)		
Post × Female	0.007^{***}		
	(0.001)		
Young × Post × Female	0.010		
	(0.008)		
Placebo Post		0.009^{***}	
		(0.001)	
Young × Placebo		0.022^{***}	
		(0.005)	
Time Trend			-0.001
			(0.001)
Young × Time Trend			-0.004
			(0.004)
Observations	888,602	459,688	888,602
R-squared	0.041	0.038	0.041

Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Dependent variable: employed (0/1). Column (5): Triple difference by gender. Column (6): Placebo test (pretend crisis in 2006). Column (7): Includes linear time trend.

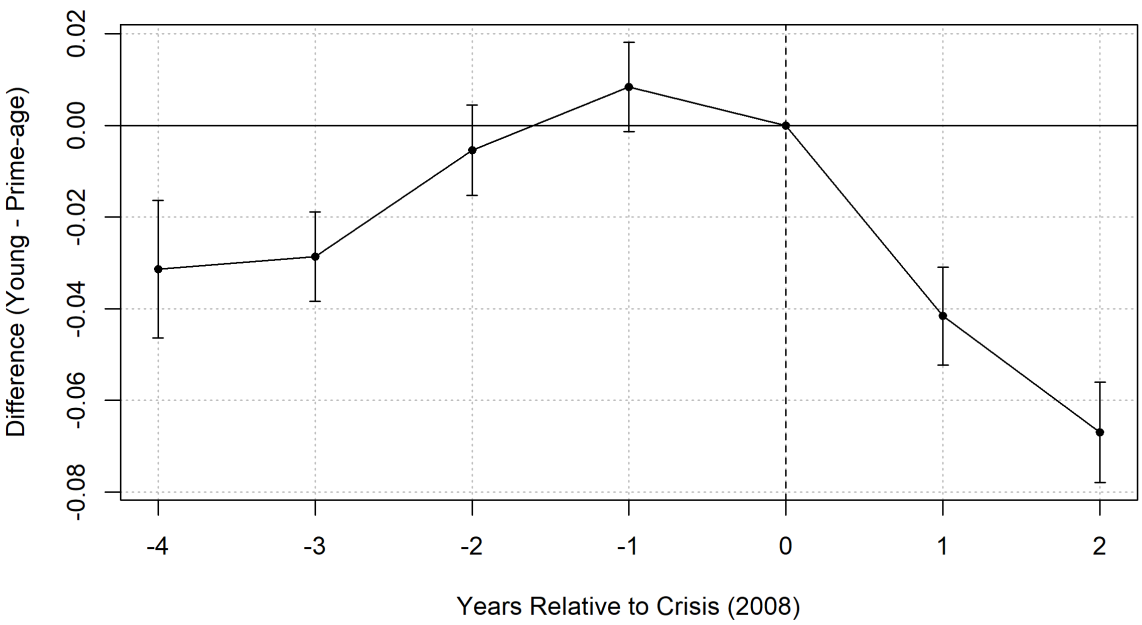

Figure 2: Event study: Youth employment gap

4.2 Event Study

Figure 2 presents event study estimates of the youth employment gap relative to 2008. The event study model is:

$$\text{Employed}_{it} = \alpha_i + \sum_{k \neq 0} \delta_k (\text{Treat}_i \times \mathbf{1}\{\text{Year}_t = k\}) + \mathbf{X}_{it}\gamma + \varepsilon_{it} \tag{9}$$

where the coefficients δ_k represent the gap between young and prime-age workers in year k relative to 2008. The pre-crisis coefficients for 2006 and 2007 are small and statistically insignificant, supporting the parallel trends assumption. Following the crisis, the gap increases drastically: young workers' employment probability declines by 4.2 percent in 2009 and 6.7 percent in 2010, showing that the crisis effect increases overtime.

5 Conclusion

This paper provides causal evidence on the impact of the 2008 Global Financial crisis on youth employment in Italy. Using a difference-in-differences framework with microdata from the EU Labour Force Survey, the crisis reduced employment probability for young workers by 3.8 percent compared to prime-age workers. This effect is robust to the inclusion of demographic controls, alternative specifications, and placebo tests.

The results also show that the crisis affected young workers in other ways as well: those

who remained employed experienced a reduction in weekly hours (0.5 hours) and an increase in part-time employment (3.6 percent). Event study estimates support the parallel pre-crisis trend and show that the negative effect increases from 2009 to 2010.

These findings have important policy implications. They emphasize the vulnerability of young workers to macroeconomic shocks.

Appendix: Variable Definitions

Table 5: Variable Definitions and Coding

Variable	Definition
employed	Indicator equal to 1 if employed (ILOSTAT = 1), 0 if unemployed (ILOSTAT = 2)
young	Indicator equal to 1 if age is between 15 and 24, 0 if age is between 30 and 50
post	Indicator equal to 1 for years 2009-2010, 0 for years 2006-2007
did	Interaction term: young × post
female	Indicator equal to 1 for female, 0 for male
age	Age in years (20, 32, or 47, due to age banding in LFS)
part_time	Indicator equal to 1 if part-time (FTPT = 2), 0 if full-time (FTPT = 1)
hwusual	Usual hours worked per week (capped at 99 for missing)

References

References

[1] Di Quirico, R. (2010). Italy and the Global Economic Crisis. *Bulletin of Italian Politics*, 2(2), 3–19.

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